TORSION FREENESS IS DETECTED ON LOCALISATIONS AT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Torsion element and torsion free module over a domain. definition at the start of the section, source lines 4781-4792. Supplies the condition that is being tested locally.
- A localisation of a torsion free module is torsion free. the statement counted by the sibling card S1-algebra-localize-torsion at 4813-4822, cited in the hint at proof lines 4859-4860. Gives the direction from torsion freeness of M to torsion freeness of its localisations.
- A module is zero if and only if all its localisations at maximal ideals are zero. the author cites Algebra, Lemma algebra-lemma-characterize-zero-local in the hint at proof line 4860. Converts the local vanishing of the annihilator of a putative torsion element into its vanishing over the ring.

Source reference. Original lemma 0AUT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13661–13661. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AUR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13659–13659. Navigation pointer to the statement of the same section that a localisation of a torsion free module is torsion free, which the author's hint invokes; nothing here re-proves it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Navigation pointer into the Algebra chapter for the criterion that a module vanishes when all its localisations at maximal ideals do; the author names it in the hint and it is a standard input.

Lemma 0.2. *Let R be a domain. Let M be an R-module. Then M is torsion free if and only if $M_{\mathfrak{m}}$ is a torsion free $R_{\mathfrak{m}}$-module for all maximal ideals $\mathfrak{m}$ of R.*

Proof. Omitted. Hint: Use Lemma 0AUR and Algebra, Lemma 00HN. □